



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)



Project Synopsis

Batch No: F7

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USN	Name	Email-ID	Mobile Number
4NI23CI075	PRACHI GOYAL	2023ci_prachigoyal_b@nie.ac.in	8571866893
4NI23CI108	SRI VARSHINI D	2023ci_srivarshinid_b@nie.ac.in	8088293876
4NI23CI116	TANISH ARORA	2023ci_tanisharora_b@nie.ac.in	9079202906

Project Title: Offline AI-Based Disaster Management and Emergency Response System

Introduction:

Natural disasters such as earthquakes, floods, cyclones, and landslides frequently disrupt communication infrastructure, making it difficult for victims to contact rescue teams during emergencies. The failure of mobile networks and internet services significantly delays rescue operations, leading to increased casualties and inefficient resource allocation. Existing disaster management systems rely heavily on internet connectivity, making them ineffective in such situations.

The proposed system addresses this challenge by enabling offline emergency communication using Bluetooth and Wi-Fi Direct. It incorporates Artificial Intelligence to classify and prioritize emergency requests, ensuring that critical cases receive immediate attention. Additionally, the system provides offline maps, a centralized rescue dashboard, and Computer Vision-based damage detection to assist rescue teams in making faster and more informed decisions during disaster response.

Objectives:

- To develop an offline emergency communication system using Bluetooth and Wi-Fi Direct.
- To enable victims to send SOS requests without internet connectivity.
- To classify and prioritize emergency requests using AI-based Natural Language Processing.
- To provide a centralized dashboard for rescue teams.
- To integrate offline maps for locating victims, hospitals, shelters, and rescue teams.
- To perform disaster damage detection using Computer Vision.
- To improve rescue response time and resource allocation during disasters.
- To improve the overall efficiency and coordination of disaster response operations through intelligent decision support.

Software Requirements:

- **OS:** Windows 11 / Ubuntu 22.04
- **Frontend:** React.js / Tailwind CSS
- **Backend:** FastAPI
- **Database:** SQLite (Offline) / PostgreSQL
- **AI/ML:** Python / Scikit-learn / PyTorch / Transformers / YOLOv8

- **Maps:** OpenStreetMap / Leaflet
- **Development Tools:** VS Code / Android Studio / Git & GitHub / Postman

Hardware Requirement:

- **Minimum:** Intel Core i5 / 8 GB RAM / 256 GB SSD / Android Smartphone / GPS-enabled Device
- **Recommended:** Intel Core i7 / Ryzen 7 / 16 GB RAM / NVIDIA GPU / Drone Images

Abstract:

Natural disasters often result in communication failures, delaying rescue operations and increasing casualties. Existing disaster management systems primarily depend on internet connectivity, making them ineffective when communication infrastructure is damaged.

This project proposes an Offline AI-Based Disaster Management and Emergency Response System that enables victims to send SOS messages using Bluetooth and Wi-Fi Direct without relying on mobile networks or internet services. Artificial Intelligence techniques are employed to classify emergency messages and assign priority levels, allowing rescue teams to respond more efficiently. The system also includes a rescue dashboard for monitoring emergency requests, offline maps for navigation, and a Computer Vision module that detects damaged buildings from drone imagery. By integrating offline communication, AI, and Computer Vision, the proposed system aims to enhance disaster response, reduce rescue time, and improve emergency management.

Literature Survey (Minimum 5 recent research Journals indexed in Scopus/Web of Science):

- **RescUE: A Resilient and Secure Device-to-Device Communication Framework for Emergencies (IEEE, 2021)**
<https://ieeexplore.ieee.org/document/9252867>
- **Bluemergency: Mediating Post-disaster Communication Systems using IoT and Bluetooth Mesh (2019)**
<https://arxiv.org/abs/1909.08094>
- **Using Artificial Intelligence to Identify Emergency Messages on social media During a Natural Disaster: A Deep Learning Approach (International Journal of Information Management Data Insights, 2023)**
<https://www.sciencedirect.com/science/article/pii/S2667096823000113>
- **BERT-Based Chinese Text Classification for Emergency Management with a Novel Loss Function (Applied Intelligence, 2022)**
<https://doi.org/10.1007/s10489-022-03946-x>
- **Real-Time Ground-Level Building Damage Detection Based on Lightweight and Accurate YOLOv5 Using Terrestrial Images (Remote Sensing, 2022)**
<https://doi.org/10.3390/rs14122763>
- **Large-Scale Building Damage Assessment Using a Novel Hierarchical Transformer Architecture on Satellite Images (Computer-Aided Civil and Infrastructure Engineering, 2023)**
<https://doi.org/10.1111/mice.12981>

Existing System & its Drawback:

Current disaster management applications such as emergency reporting systems rely primarily on internet connectivity and centralized cloud infrastructure for communication and coordination.

Drawbacks

- Cannot function when internet connectivity is unavailable.
- Delayed communication during disasters.

- Manual prioritization of emergency requests.
- Limited coordination among rescue teams.
- Lack of intelligent emergency classification.
- Slow damage assessment.
- No offline navigation support.

Proposed System & Advantages:

The proposed system provides an AI-powered offline disaster management platform where victims can send SOS messages using Bluetooth and Wi-Fi Direct. Emergency requests are analysed using AI-based Natural Language Processing to determine their priority. Rescue teams monitor requests through a centralized dashboard, while offline maps assist in navigation. Computer Vision models analyse drone images to detect damaged buildings for faster disaster assessment.

Advantages

- Works without internet connectivity.
- Faster emergency communication.
- AI-based emergency prioritization.
- Offline maps for disaster navigation.
- Faster damage assessment using Computer Vision.
- Improved rescue coordination.
- Better resource utilization.
- Scalable and cost-effective.

Guide Details:

Name: Dr. B.R. Vatsala

Designation: Associate Professor

Mobile Number: 99456 16978

College Address: No 50, Koorgalli Village, Hootagalli Industrial Area, Next to BEML, Mysore, Karnataka, 570018